

Hong-Bo Li — Curriculum Vitae

CONTACT INFORMATION	2906S, Science Teaching Building No. 2, Department of Astronomy. Peking University, 5 Yiheyuan Road, Haidian District Beijing 100871, P. R. China	email: lihb2020@pku.edu.cn
EDUCATION	Post-doctor, Physics , Peking University, Beijing, China Advisor: Prof. Renxin Xu	July 2024-2026
EDUCATION	Ph.D., Physics , Peking University, Beijing, China Thesis Advisor: Prof. Lijing Shao and Prof. Renxin Xu Thesis Title: <i>The asteroseismology of compact stars</i>	June 2024
	M.S., Physics , Lanzhou University, Lanzhou, Gansu Province, China Senior Dissertation Advisors: Prof. Yongqiang Wang Dissertation Title: <i>Numerical study of boson stars</i> Excellent Master's Thesis of Gansu Province in 2020	July 2020
	B.S., Physics , Shandong Normal University, Jinan, Shandong Province, China Senior Dissertation Advisors: Prof. Yanhui Zhang	July 2017
RESEARCH INTERESTS	Gravitational waves from asymmetrically deformed spinning pulsars (1) Astrophysics with continuous gravitational waves; (2) Detecting gravitational waves from mountains on pulsar in the advanced detector era. Dynamical properties of magnetic polar regions in X-ray pulsar (1) Structure and observable properties of accretion columns in X-ray pulsars; (2) The thermal evolution of transiently accreting pulsars. Astrophysical radiation hydrodynamics (1) Dynamics of the flows accreting onto pulsars; (2) Super-Eddington accretion discs around pulsars.	
FUNDS	1. National Natural Science Foundation of China, 12447148	2025
	2. China Postdoctoral Science Foundation, 2024M760081	2025
HONORS AND AWARDS	The Third Prize for Zhongshengbiao Academic Forum , Peking University	May 2023
	The Best Poster Presentation , Australia-China Workshop on Astrophysics	October 2022
	Excellent Teaching Assistant Award , Peking University	2021–2022
	Excellent Master's Thesis of Gansu Province , Lanzhou University	2020
	National Scholarship , Lanzhou University	2018–2019

TEACHING
EXPERIENCE

Teaching Assistant, Peking University

Theoretical Mechanics (A), Excellent Teaching Assistant Award**Fall 2021**

Atomic Physics

Fall 2022

Astrophysics

Spring 2024CO-ADVISED
STUDENTS

Undergraduate Students, co-advised with Prof. Lijing Shao

Yukun Liu, Peking University**2022-Present***Effects from Dark Matter Halos on X-ray Pulsar Pulse Profiles***Chenghui Zhao**, NanJing University**2023-Present***Effects of Crystallization on Oscillation Modes of White Dwarfs***Ziyuan Nie**, University of Oxford**2024-Present***Torsional oscillations of neutron stars in scalar-tensor gravity*

COMPUTER SKILLS

Proficient in MATHEMATICA and Python. Markup languages: L^AT_EX.

Organizer and chair of conference/group meeting

Local Organizer of the **SKA Pulsar Science Symposium (2025)****July 2025**Local Organizer of the **SKA Pulsar Science Symposium (2022)****August 2022**Chair of the **KIAAGRAVITY GROUP MEETING****2021-2022**SUBMITTED
PUBLICATIONS

24. **H.-B. Li**, Yacheng Kang Ren-Xin Xu, (2025), *Thermal Deformations in Super-Eddington Magnetized Neutron Stars: Implications for Continuous Gravitational-Wave Detectability*, submitted to ApJ Letter; [[arXiv:2510.10443](#)]

REFEREED
PUBLICATIONS

23. **H.-B. Li**, Shi-Jie Gao, Xiang-Dong Li, Ren-Xin Xu, (2026), *Polar Mounds on Strangeon Stars: the Neutrino Emission from Ultraluminous X-ray Pulsars*, **Mon. Not. R. Astron. Soc.** **546**, [stag241](#), [[arXiv:2509.13732](#)].
22. Jinghao Zhang, Yacheng Kang, Jiahang Zhong, **H.-B. Li**, Liang-Duan Liu, Yun-Wei Yu, Lijing Shao, (2026), *Lightcurve Features of Magnetar-Powered Superluminous Supernovae with Gravitational-Wave Emission and High-Energy Leakage*, **RAA**, **26**, [035013](#), [[arXiv:2508.18940](#)].
21. Chenghui Zhao, **H.-B. Li**, L. Shao (2025), *Effects of Crystallization on Oscillation Modes of White Dwarfs*, accepted by Progress in Astronomy.
20. Hao Sun, Jia-Xing Niu **H.-B. Li**, C.-J. Xia, Enping Zhou, Yiqiu Ma, Ying-Xun Zhang, (2025), *Effects of inner crusts on g-mode oscillations in neutron stars*, **Phys. Rev. D**, **111**, [103019](#), [[arXiv:2501.07188](#)]
19. **H.-B. Li**, L. Shao, C.-J. Xia, R.-X. Xu, (2025), *Continuous gravitational waves from thermal mountains on accreting neutron stars: effect of the nuclear pasta phase*, **Astrophys. J.**, **980**, [144](#), [[arXiv:2411.11075](#)]
18. Yukun Liu, **H.-B. Li**, Yong Gao, L. Shao, Zexin Hu, (2024), *Effects from Dark Matter Halos on X-ray Pulsar Pulse Profiles*, **Phys. Rev. D**, **110**, [083018](#), [[arXiv:2408.04425](#)]
17. H.-S. You, H. Sun, **H.-B. Li**, C.-J. Xia, R.-X. Xu, (2024), *Compact dwarfs made of light-quark nuggets*, **Phys. Rev. D**, **110**, [054041](#), [[arXiv:2406.00613](#)]
16. R.-P. Lu, et al., (2024), *Quakes: from the Earth to Stars*, **SCIENTIA SINICA Physica Mechanica Astronomica**, **54**, [289501](#)

15. **H.-B. Li**, Y. Gao, L. Shao, R.-X. Xu, (2024), **Review paper: Asteroseismology of Compact Stars**, *Universe*, **10**, 157
14. W.-Y. Wang, Y.-P. Yang, **H.-B. Li**, J.-F. Liu, R.-X. Xu, (2024), *Narrow spectra of repeating fast radio burst: A magnetospheric origin*, *Astron. Astrophys.*, **685**, A87 [arXiv:2311.13114]
13. **H.-B. Li**, Y.-C. Kang, Z.-X. Hu, C.-J. Xia, L. Shao, R.-X. Xu, (2023), *Quasi-periodic oscillations during magnetar giant flares in the strangeon star model*, *Mon. Not. R. Astron. Soc.* **527**, 855 [arXiv:2309.09847]
12. C. Zhang, Y.-D. Luo, **H.-B. Li**, L. Shao, R.-X. Xu, (2023), *Radial and Non-Radial Oscillations of Inverted Hybrid Stars*, *Phys. Rev. D*, **109**, 063020, [arXiv:2306.08234]
11. C.-J. Xia, Y.-F. Huang, **H.-B. Li**, L. Shao, R.-X. Xu, (2023), *Ultra low-mass and small-radius white dwarfs made of heavy elements*, *Frontiers in Astronomy and Space Sciences* **10**, 1334642 [arXiv:2305.05847]
10. **H.-B. Li**, Y. Gao, L. Shao, R.-X. Xu, (2023), *g-mode of neutron stars in pseudo-Newtonian gravity*, *Phys. Rev. D*, **108**, 064005 [arXiv:2302.03856].
9. **H.-B. Li**, Y. Gao, L. Shao, R.-X. Xu, R. Xu, (2022) *Oscillation modes and gravitational waves from strangeon stars*, *Mon. Not. R. Astron. Soc.* **516**, 6172 [arXiv:2206.09407].
8. **H.-B. Li**, L. Shao, R.-X. Xu, (2021) *Gravitational wave bursts from core-collapse supernovae*, *Progress in Astronomy.*, **4**, 39.
7. **H.-B. Li**, Y.-B. Zeng, Y. Song, Y.-Q. Wang, (2021), *Self-interacting multistate boson stars*, *JHEP*, **04**, 042 [arXiv:2006.11281].
6. Y.-Q. Wang, **H.-B. Li**, Y.-X. Liu, Y. Zhong, (2021), *Excited states of holographic superconductors with backreaction*, *Eur. Phys. J. C*, **81**, 628 [arXiv:1911.04475].
5. **H.-B. Li**, S. Sun, T.-T. Hu, Y.-Q. Wang, (2020), *Rotating multistate boson stars*, *Phys. Rev. D*, **101**, 044017, [arXiv:1906.00420].
4. T.-T. Hu, S. Sun, **H.-B. Li**, Y.-Q. Wang, (2020), *Deforming charged black holes with dipolar differential rotation boundary*, *Eur. Phys. J. C*, **80**, 599 [arXiv:1909.01628].
3. **H.-B. Li**, T.-T. Hu, S. Sun, Y.-Q. Wang, (2019), *Deforming black holes with even multipolar differential rotation boundary*, *JHEP*, **06**, 126 [arXiv:1903.11967].
2. S. Sun, T.-T. Hu, **H.-B. Li**, Y.-Q. Wang, (2019), *Deforming black holes with odd multipolar differential rotation boundary*, *Phys. Rev. D*, **100**, 084063 [arXiv:1906.06183].
1. T.-T. Hu, S. Sun, **H.-B. Li**, Y.-Q. Wang, (2019), *Weak cosmic censorship in Born-Infeld electrodynamics and bound on charge-to-mass ratio*, *Eur. Phys. J. C*, **80**, 147 [arXiv:1906.00235].

INVITED TALKS

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|---|------------|
| 3. Henan, Henan academy of sciences | June 2025 |
| 2. Gansu, Langzhou University, School of Physics Science and Technology | July 2024 |
| 1. Jiangsu, Yangzhou University, School of Physics Science and Technology | April 2023 |

CONTRIBUTED
TALKS AND POSTERS

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|---|---------------|
| 21. CNAPS2025 | December 2025 |
| 20. The 7th Symposium on Multiband Studies of X-ray binaries | December 2025 |
| 19. 2025 Academic Annual Meeting of the Chinese Astronomical Society | October 2025 |
| 18. Binary Stars in a New Era | August 2025 |
| 17. ICESUN Summer School 2025: Binary Stars and Compact Objects | August 2025 |
| 16. FAST/Future Pulsar Symposium 14 | July 2025 |
| 15. SKA Pulsar Science Symposium 2025 | July 2025 |
| 14. The Era When Pulsars Meet FRBs | May 2025 |

13. [The 2025 Academic Annual Conference of the Gravity and Relativistic Astrophysics](#) April 2025
12. [The International Pulsar Symposium 2024](#) December 2024
11. [The 6th Symposium on Multiband Studies of X-ray binaries](#) November 2024
10. [Dialog at the Dream Field: Supranuclear Matter](#) May 2024
9. Asia-Pacific Regional IAU Meeting 2023 (*Japan*) August 2023
8. SKA Pulsar Science Symposium 2023 July 2023
7. FAST/Future Pulsar Symposium 12 July 2023
6. Quakes: From the Earth to Stars May 2023
5. Gravitation and Relativistic Astrophysics, Chinese Physical Society April 2023
4. The 6th Northwest Area Theoretical Physics Frontier Academic Seminar November 2022
3. ACAMAR 8: Australia-China Workshop on Astrophysics (*online*) October 2022
2. The 9th KAGRA International Workshop (*online*) June 2022
1. Gravitation and Relativistic Astrophysics, Chinese Physical Society July 2019

PHD AND
POST-DOCTOR
ADVISORS

Lijing Shao, Assistant Professor of Kavli Institute for Astronomy and Astrophysics, Peking University
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